

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I Geferlin Boffy R (192571080) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

1. Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

Schema

A **Schema** is the logical design or blueprint of a database. It defines the structure of the database, including tables, attributes, relationships, constraints, and data types. A schema is created before data is stored and usually remains unchanged unless the database design is modified.

Example:

Student(Student_ID, Name, Department)

This defines the structure of the Student table.

Instance

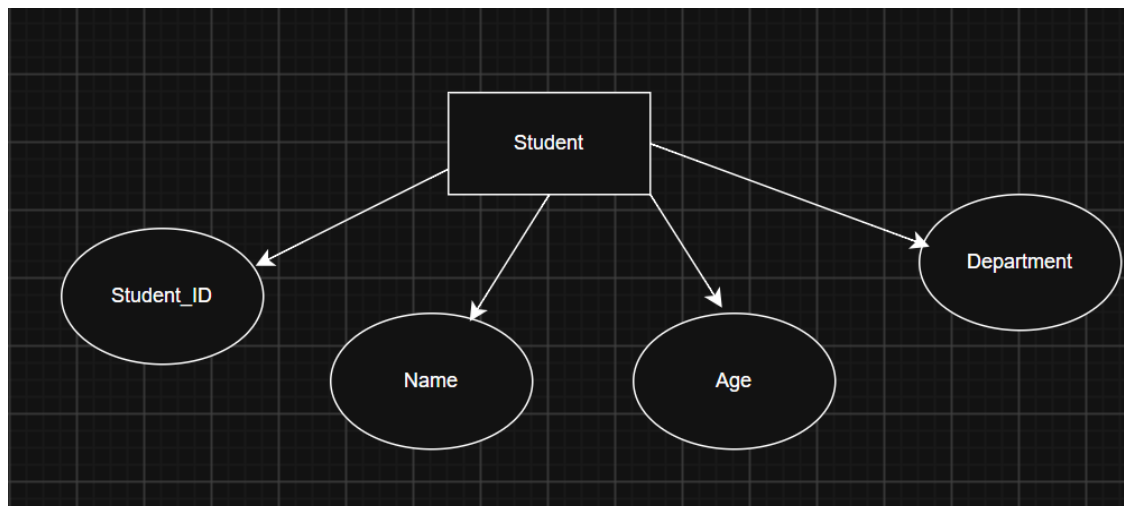
An **Instance** is the actual data stored in the database at a particular moment. Unlike the schema, the instance changes frequently as records are inserted, updated, or deleted.

Example:

Student_ID Name Department

101 Rahul CSE

102 Priya AI



Difference between Schema and Instance

Schema	Instance
Blueprint of the database	Actual data stored
Changes rarely	Changes frequently
Defines structure	Represents current data

Types of Schema

Logical Schema

- Defines tables, attributes, relationships, and constraints.
- Independent of physical storage.
- Used by database designers.

Physical Schema

- Describes how data is stored on disk.
- Includes files, indexes, and storage techniques.
- Managed by the DBMS.

View Schema

- Displays only the required data to users.
- Hides unnecessary information.
- Provides security and simplicity.

2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

A **Bank Management System** is used to store and manage information about customers, accounts, employees, and transactions. The ER diagram helps represent the database structure by identifying entities, attributes, and relationships.

Entities

- Customer
- Account
- Transaction
- Employee

Attributes

Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone Number

Account

- Account_No (Primary Key)
- Account_Type
- Balance

Transaction

- Transaction_ID (Primary Key)
- Date

- Amount
- Transaction_Type

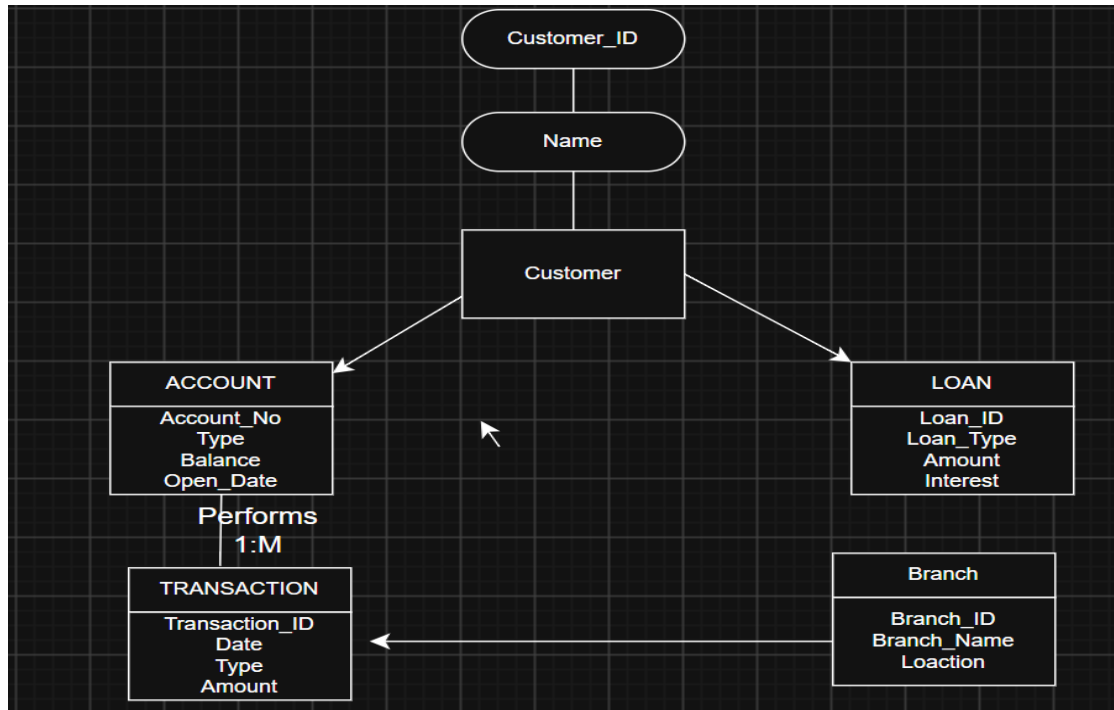
Employee

- Employee_ID (Primary Key)
- Name
- Designation

Relationships

- One customer can own one or more accounts.
- One account can have many transactions.
- Employees manage customer accounts.
- Each transaction belongs to one account.

ER DIAGRAM



Advantages

- Organizes customer information efficiently.
- Reduces data redundancy.
- Maintains data integrity.
- Simplifies banking operations.

3. Explain the Extended ER (EER) Model. Describe the concepts of Generalization, Specialization, and Aggregation.

The **Extended Entity Relationship (EER) Model** is an advanced version of the ER Model. It provides additional features to represent complex real-world database systems. The EER model introduces concepts such as inheritance, generalization, specialization, and aggregation.

Generalization

Generalization is the process of combining two or more similar lower-level entities into a single higher-level entity called a **Superclass**. The superclass contains common attributes shared by all subclasses.

Example:

- Student and Faculty are combined into Person.

Specialization

Specialization is the process of dividing a higher-level entity into smaller, more specific entities called **Subclasses**. Each subclass inherits the attributes of the superclass and also contains its own specific attributes.

Example:

- Person is divided into Student and Faculty.

Aggregation

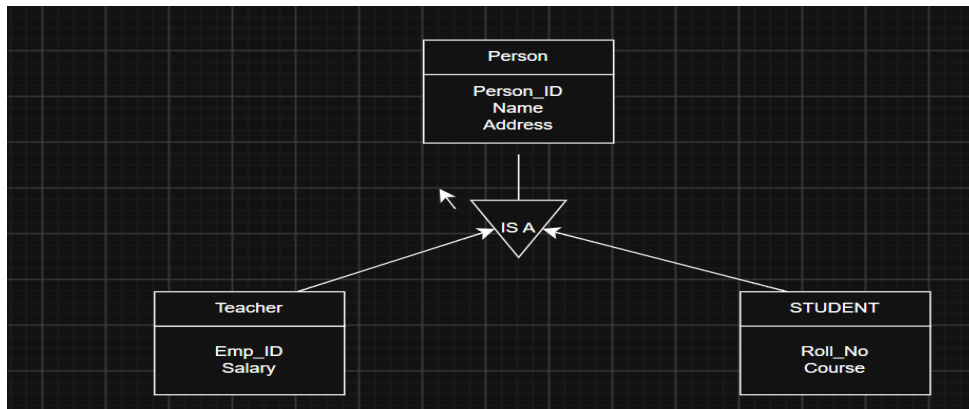
Aggregation is used when a relationship between entities needs to be treated as a higher-level entity. It helps represent complex relationships involving multiple entities.

Example:

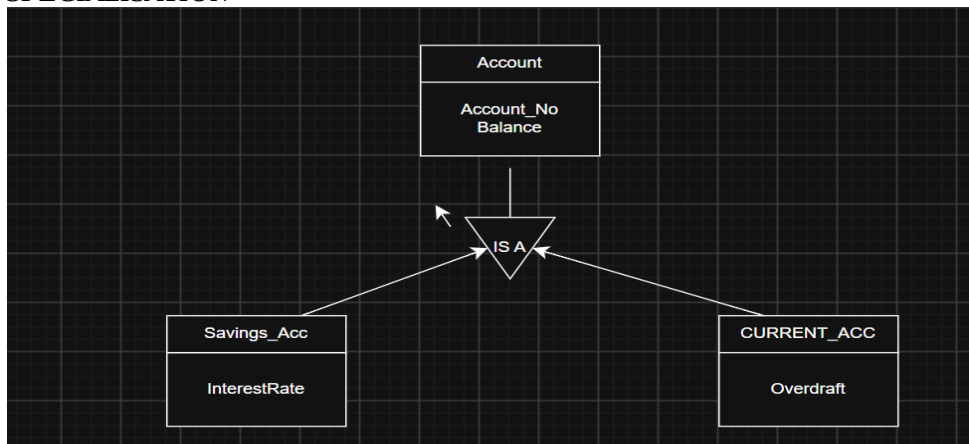
- Employee works on Project.
- The relationship "Works On" is considered as an Assignment entity.

ER DIAGRAM

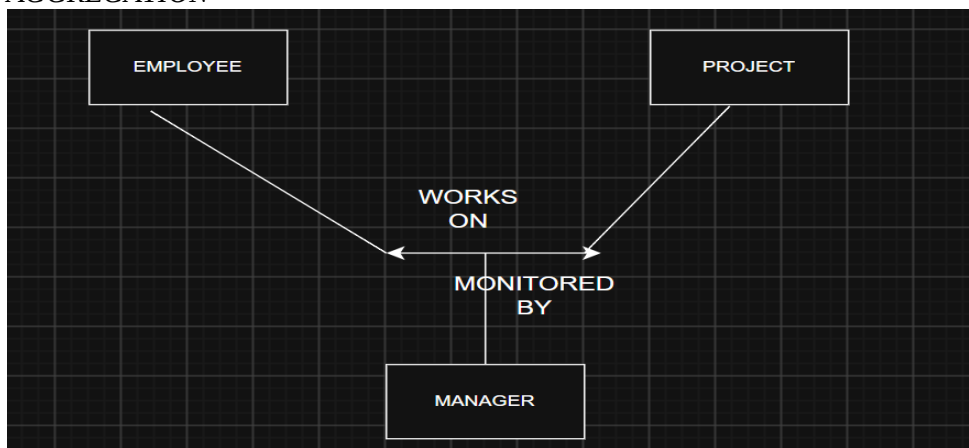
GENERALISATION



SPECIALISATION



AGGREGATION



Advantages of EER Model

- Represents complex databases effectively.
- Reduces duplicate data.
- Supports inheritance.
- Improves database organization.
- Makes future modifications easier.

4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

A **Hospital Management System** stores and manages information related to doctors, patients, appointments, treatments, and medical records. The EER model is used to represent inheritance and weak entities.

Superclass

Person

- Person_ID
- Name
- Age
- Gender

Subclasses**Doctor**

- Doctor_ID
- Specialization
- Experience

Patient

- Patient_ID
- Disease
- Blood Group

Both Doctor and Patient inherit the common attributes of Person.

Weak Entity**Medical Record**

- Record_Number (Partial Key)
- Diagnosis
- Treatment
- Prescription

The Medical Record cannot exist without a Patient, so it is considered a **Weak Entity**.

Relationships

- Doctor treats Patient.
- Patient has Medical Record.
- Person is the Superclass.
- Doctor and Patient are Subclasses.

Advantages

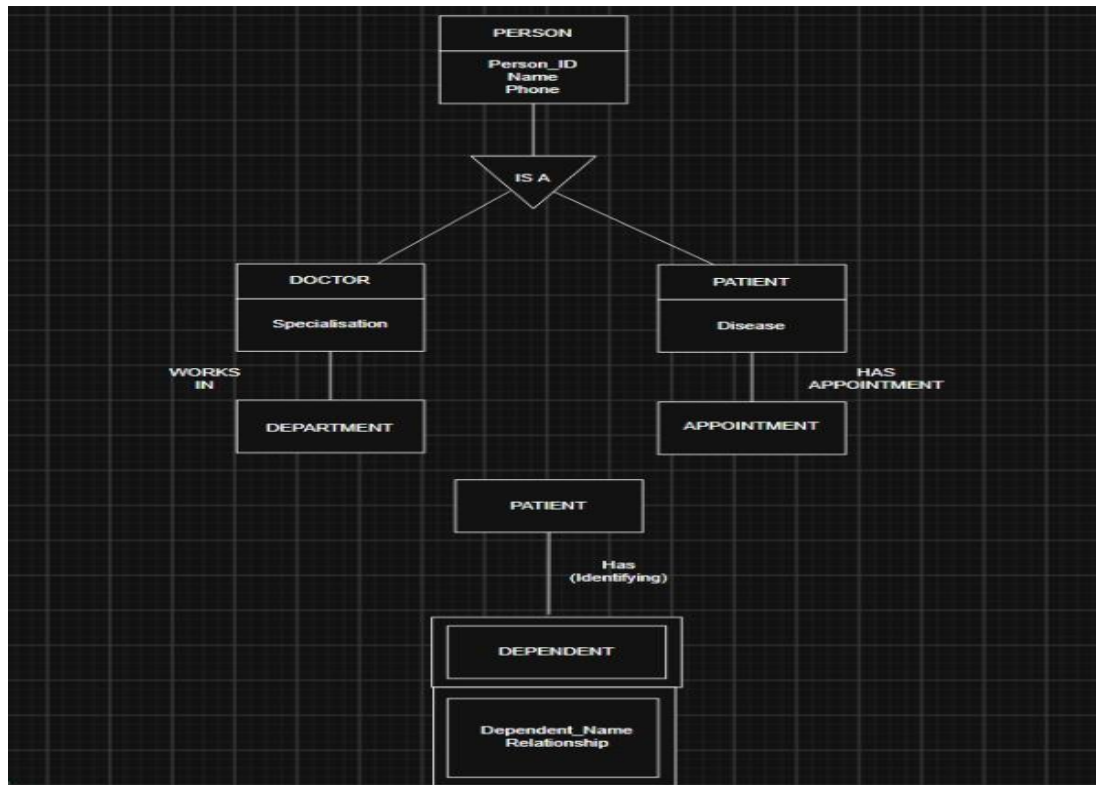
- Reduces duplicate information.
- Improves database organization.
- Supports inheritance.
- Represents dependent entities clearly.

5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

A **Database Administrator (DBA)** is a professional responsible for managing, maintaining, securing, and monitoring databases. The DBA ensures that the database operates efficiently, securely, and is always available for users.

Responsibilities of a DBA

- Install and configure database software.
- Create and maintain databases.
- Manage user accounts and permissions.
- Perform regular database backup and recovery.
- Monitor database performance.
- Optimize SQL queries for better speed.
- Ensure database security and prevent unauthorized access.
- Maintain data integrity and consistency.
- Troubleshoot database issues.
- Perform software updates and maintenance.



Importance of a DBA

- Protects important organizational data.
- Ensures high database performance.
- Prevents data loss through backups.
- Provides secure access to authorized users.
- Supports smooth business operations.
- Maintains data accuracy and reliability.